

**Applying Exponent Laws***Independent Set 3.05*   **ANSWER KEY***Six questions, on your own, in silence. Show the step that does the work.*

1. Rewrite with a positive exponent.  $\left(\frac{3}{4}\right)^{-2}$

$$\left(\frac{4}{3}\right)^2$$

2. Find the value.  $\left(\frac{2}{5}\right)^{-2}$

$$\frac{25}{4}$$

*Flip, then square.*

3. Rewrite with a positive exponent.  $\left(-\frac{5}{6}\right)^{-3}$

$$\left(-\frac{6}{5}\right)^3$$

*The base keeps its sign through the flip.*

4. Find the value.  $\left(\frac{1}{2}\right)^{-4}$

$$16$$

 *$(2/1)^4$ .*

5. Find the value.  $\left(\frac{3}{2}\right)^{-1} \cdot \left(\frac{3}{2}\right)^3$

$$\left(\frac{3}{2}\right)^2 = \frac{9}{4}$$

 *$-1 + 3 = 2$ .*

6. Find the value of  $n$ .  $\left(\frac{4}{7}\right)^n = \left(\frac{7}{4}\right)^3$

$$n = -3$$

*Flipping the base flips the exponent's sign.*